Slide Set #9

Linear model with predermined regressors and its OLS estimation

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Notes

Assumptions

Assumption (B1): linearity

The process $\{y_i, \boldsymbol{x}_i\}$ is generated by

$$y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i, \qquad \text{for } i = 1, 2, \ldots, n$$

where: $\boldsymbol{x}_i$ is the K-dimensional vector of explanatory variables (regressors), $\boldsymbol{\beta}$ is the K-dimensional coefficient vector, ε_i is the unobservable error term.

This the same functional specification that we had in the CLRM

Notes

Assumption (B2): ergodic stationarity

The $(K+1)$-dimensional vector process $\{y_i, \boldsymbol{x}_i\}$ is jointly stationary and ergodic.

Observe

- iid sequences are trivially ergodic stationary
- $\varepsilon_i = y_i - \boldsymbol{x}_i'\boldsymbol{\beta}$ is ergodic stationary

Stationarity and heteroscedasticity

- since $\{\varepsilon_i\}$ is stationary, $\mathrm{E}[\varepsilon_i^2]$ cannot depend on i, therefore *unconditional heteroscedasticity is not allowed*
- (B2) does not restricts $\mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i]$ to not depend on i. Therefore *conditional heteroscedasticity is allowed*

A stationary process needs to be unconditionally homoskedastic

Notes

Remark on conditional homo/heteroscedasticity

Conditional homoskedasticity $\implies$ unconditional homoskedasticity

it easy to see that if $\mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i] = c$, then $\mathrm{E}[\varepsilon_i^2] = \mathrm{E}[\mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i]] = \mathrm{E}[c] = c$

The converse is not necessarily true

Assume $\{\varepsilon_i\}$ is *unconditionally homoskedastic*, that is $\mathrm{E}[\varepsilon_i^2] = c$. Assume also $\varepsilon_i = u_i h(\boldsymbol{x}_i)$, where u_i and x_i are independent. Wlog suppose $\mathrm{E}[u_i^2] = 1$

$$\begin{aligned}
\mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i] &= \mathrm{E}[u_i^2 h(\boldsymbol{x}_i)^2 \,|\, \boldsymbol{x}_i] \\
&= h(\boldsymbol{x}_i)^2\, \mathrm{E}[u_i^2 \,|\, \boldsymbol{x}_i] \quad \text{(pull out what's known)} \\
&= h(\boldsymbol{x}_i)^2\, \mathrm{E}[u_i^2] \quad \text{(because of independence)} \\
&= h(\boldsymbol{x}_i)^2
\end{aligned}$$

$\mathrm{E}[\varepsilon_i^2 \,|\, \boldsymbol{x}_i]$ depends on i, therefore $\{\varepsilon_i\}$ is conditionally heteroscedastic

This should convince you that a stationary process that must be unconditionally homoskedastic, may still be conditionally heteroskedastic

Notes

Assumption (B3): predetermined regressors/orthogonality

All regressors are *predetermined*, in the sense that they are *orthogonal* to the *contemporaneous* error:

$$\mathrm{E}[x_{ik}\varepsilon_i] = 0 \qquad \text{for all} \quad i = 1, 2, \ldots, n \quad \text{and} \quad k = 1, 2, \ldots, K$$

Denote

$$\boldsymbol{g}_i := \boldsymbol{x}_i\varepsilon_i = \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta}),$$

then (B3) is equivalent to $\mathrm{E}[\boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})] = \mathrm{E}[\boldsymbol{g}_i] = 0$

Remark

$\mathrm{E}[\varepsilon_i|\boldsymbol{x}_i] = 0 \implies \mathrm{E}[\boldsymbol{x}_i\varepsilon_i] = \mathbf{0}$, therefore (B3) is weaker than $\mathrm{E}[\varepsilon_i|\boldsymbol{x}_i] = 0$

Notes

Conditional mean interpretation

Under the stronger $\mathrm{E}[\varepsilon_i|\boldsymbol{x}_i] = 0$, the *linear conditional mean regression function* interpretation of (B1) is obtained: $\mathrm{E}[y_i \,|\, \boldsymbol{x}_i] = \boldsymbol{x}_i'\boldsymbol{\beta}$

Predeterminedness vs strict exogeneity

- Recall the *strict exogeneity assumption* (A2): for any of the kth -regressor $\mathrm{E}[x_{jk}\varepsilon_i] = 0$ for all i, j
- under (A2) the current error ε_i cannot be correlated with paste/future regressors. For example, (A2) does not allow that ε_{i-1} correlates with x_{ik} for some k
- (B3) is weaker than strict exogeneity, because it only imposes orthogonality on contemporaneous relationships between the error and the regressors

Notes

Assumption (B4): rank condition

The $(K \times K)$ matrix $\boldsymbol{\Sigma}_{\boldsymbol{xx}} = \mathrm{E}[\boldsymbol{x}_i \boldsymbol{x}_i']$ is non singular (and thus finite)

Inverse of $\boldsymbol{X}'\boldsymbol{X}$

$\boldsymbol{S}_{\boldsymbol{xx}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i' = \frac{1}{n}\boldsymbol{X}'\boldsymbol{X}$. By the ergodic $\boldsymbol{S}_{\boldsymbol{xx}} \xrightarrow{\text{as}} \mathrm{E}[\boldsymbol{x}_i \boldsymbol{x}_i']$. This is a tech requirement because we need to invert $\boldsymbol{S}_{\boldsymbol{xx}}$ all the time

Rank of $\boldsymbol{X}'\boldsymbol{X}$

if $\frac{1}{n}\boldsymbol{X}'\boldsymbol{X}$ is non singular then rank$(\boldsymbol{X}) = K$. Hence, we are somehow requiring that in large samples with probability 1 the design matrix doesn't suffer from multicollinearity

Notes

Assumption (B5): MDS dependence

The process $\{\boldsymbol{g}_i\}$ is MDS with non singular second (cross) moment $\boldsymbol{S} = \mathrm{E}[\boldsymbol{g}_i \boldsymbol{g}_i']$

(B3) establishes $\mathrm{E}[\boldsymbol{g}_i] = \boldsymbol{0}$, now (B5) adds more structure by requiring an MDS behavior (needed for CLTs).

Note that $\boldsymbol{S}$ is the covariance matrix of the process $\{\boldsymbol{g}_i\}$, because by (B3)

$$\mathrm{Cov}[\boldsymbol{g}_i] = \mathrm{E}[\boldsymbol{g}_i \boldsymbol{g}_i'] - \mathrm{E}[\boldsymbol{g}_i]\mathrm{E}[\boldsymbol{g}_i]' = \mathrm{E}[\boldsymbol{g}_i \boldsymbol{g}_i']$$

Moreover

$$\boldsymbol{S} = \mathrm{E}[\boldsymbol{g}_i \boldsymbol{g}_i'] = \mathrm{E}[(\boldsymbol{x}_i \varepsilon_i)(\boldsymbol{x}_i \varepsilon_i)'] = \mathrm{E}[\varepsilon_i^2 \boldsymbol{x}_i \boldsymbol{x}_i']$$

Therefore $\boldsymbol{S}_{kj} = \mathrm{E}[\varepsilon_i^2 x_{ik} x_{ij}]$. The previous (B5) requires that all these mixed high order moments exist.

Notes

The MDS assumption on the product process $\{\boldsymbol{g}_i\}$ is difficult to interpret.

A sufficient condition for (B5) is

$$\mathrm{E}[\varepsilon_i \mid \{\varepsilon_{i-1}, \ldots, \varepsilon_1, \boldsymbol{x}_i, \boldsymbol{x}_{i-1}, \ldots, \boldsymbol{x}_1\}] \;=\; 0 \qquad \text{(B5-ERR)}$$

If (B5-ERR) holds then $\{\varepsilon_i\}$ is MDS, and therefore $\{\varepsilon_i\}$ is serially uncorrelated (see homeworks).

Under (B5-ERR) one can again construct another meaningful interpretation of the regression function because

$$\mathrm{E}[y_i \mid \{\varepsilon_{i-1}, \ldots, \varepsilon_1, \boldsymbol{x}_i, \boldsymbol{x}_{i-1}, \ldots, \boldsymbol{x}_1\}] = \boldsymbol{x}_i'\boldsymbol{\beta}$$

Notes

Models with constant term

Assume $x_{i1} = 1$ for all i. Therefore the first element of $\boldsymbol{g}_i$ is simply $x_{i1}\varepsilon_i = \varepsilon_i$

It is easy to see that (B3) implies that for all $i = 1, 2, \ldots$

- $\mathrm{E}[\varepsilon_i] = 0$
- $\mathrm{Cov}[x_{ik}, \varepsilon_i] = 0$ for all $k \neq 1$

It can be also shown (see homeworks) that (B5) implies that the error term is a scalar MDS process, that is

$$\mathrm{E}[\varepsilon_i \,|\, \varepsilon_{i-1}, \varepsilon_{i-2}, \ldots, \varepsilon_1] = 0,$$

an therefore it is serially uncorrelated

Notes

Properties of the OLS estimator

Proposition (Consistency of the OLS)

Assume (B1)–(B4). Then $\boldsymbol{b} \xrightarrow{\text{p}} \boldsymbol{\beta}$ as $n \to \infty$.

Proof. Write the usual estimation OLS error as

$$\begin{aligned}\boldsymbol{b} - \boldsymbol{\beta} =& (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\varepsilon} \\ =& \left(\frac{1}{n}\boldsymbol{X}'\boldsymbol{X}\right)^{-1}\left(\frac{1}{n}\boldsymbol{X}'\boldsymbol{\varepsilon}\right) \\ =& \left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i\boldsymbol{x}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i\varepsilon_i\right) \\ =& \boldsymbol{S}_{\boldsymbol{xx}}^{-1}\overline{\boldsymbol{g}}\end{aligned}$$

where

$$\overline{\boldsymbol{g}} := \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{g}_i \quad \text{and} \quad \boldsymbol{g}_i := \boldsymbol{x}_i\varepsilon_i$$

Notes

By (B2) $\{\boldsymbol{x}_i\boldsymbol{x}_i'\}$ is ergodic stationary, hence $\boldsymbol{S}_{\boldsymbol{xx}} \xrightarrow{\text{as}} \boldsymbol{\Sigma}_{\boldsymbol{xx}} \implies \boldsymbol{S}_{\boldsymbol{xx}} \xrightarrow{\text{p}} \boldsymbol{\Sigma}_{\boldsymbol{xx}}$.

Moreover, by (B4) $\boldsymbol{\Sigma}_{\boldsymbol{xx}}$ is invertible, hence $\boldsymbol{S}_{\boldsymbol{xx}}^{-1} \xrightarrow{\text{p}} \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1}$ by the continuous mapping theorem.

(B2)+(B3) $\implies \overline{\boldsymbol{g}} \xrightarrow{\text{p}} \mathrm{E}[\boldsymbol{g}_i] = \boldsymbol{0}$. Applying again the continuous mapping theorem: $\boldsymbol{S}_{\boldsymbol{xx}}^{-1}\overline{\boldsymbol{g}} \xrightarrow{\text{p}} \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1}\,\mathrm{E}[\boldsymbol{g}_i] = \boldsymbol{0}$.

Therefore $\boldsymbol{b} - \boldsymbol{\beta} \xrightarrow{\text{p}} \boldsymbol{0}$, that is $\boldsymbol{b} \xrightarrow{\text{p}} \boldsymbol{\beta}$

Notes

Proposition (Asymptotic normality)

Assume (B1)–(B5). Then for $n \to \infty$

$$\sqrt{n}(\boldsymbol{b} - \boldsymbol{\beta}) \xrightarrow{\mathsf{d}} \mathcal{N}(\mathbf{0}, \mathrm{AVar}(\boldsymbol{b})), \quad \text{where} \quad \mathrm{AVar}(\boldsymbol{b}) = \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1} \boldsymbol{S} \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1}$$

Observe: $\mathrm{AVar}(\boldsymbol{b})$ is the asymptotic variance of $\sqrt{n}(\boldsymbol{b} - \boldsymbol{\beta})$

Proof. $(\boldsymbol{b} - \boldsymbol{\beta}) = \boldsymbol{S}_{\boldsymbol{xx}}^{-1}(\overline{\boldsymbol{g}})$, then

$$\sqrt{n}(\boldsymbol{b} - \boldsymbol{\beta}) = \boldsymbol{S}_{\boldsymbol{xx}}^{-1} \sqrt{n}(\overline{\boldsymbol{g}})$$

By (B5) $\sqrt{n}\overline{\boldsymbol{g}} \xrightarrow{\mathsf{d}} \mathcal{N}(\mathbf{0}, \boldsymbol{S})$.

(B2)+(B4)+continuous mapping theorem $\implies \boldsymbol{S}_{\boldsymbol{xx}}^{-1} \xrightarrow{\mathsf{p}} \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1}$.
By Slutzky Theorem

$$\sqrt{n}(\boldsymbol{b} - \boldsymbol{\beta}) \xrightarrow{\mathsf{d}} \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1} \mathcal{N}(\mathbf{0}, \boldsymbol{S}) \quad = \quad \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1} \boldsymbol{S} (\boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1})'),$$

$\boldsymbol{\Sigma}_{\boldsymbol{xx}}$ and $\boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1}$ are symmetric, therefore $\mathrm{AVar}(\boldsymbol{b}) = \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1} \boldsymbol{S} \boldsymbol{\Sigma}_{\boldsymbol{xx}}^{-1}$, which proves the result. ■

Notes

Error variance

Let $e_i = y_i - \boldsymbol{x}_i' \boldsymbol{b}$ be the residual. An asymtotically unbiased and consistent estimator for the variance of the error term is

$$s^2 := \frac{1}{n - K} \sum_{i=1}^{n} e_i^2$$

Proposition (Consistency of s^2)

Assume (B1)–(B4), then $s^2 \xrightarrow{\mathsf{p}} \mathrm{E}[\varepsilon_i^2]$ as $n \to \infty$.

Proof. We can write e_i in terms of the estimation error:

$$\begin{aligned} e_i =& y_i - \boldsymbol{x}_i' \boldsymbol{b} \\ =& y_i - \boldsymbol{x}_i' \boldsymbol{\beta} - \boldsymbol{x}_i' \boldsymbol{b} + \boldsymbol{x}_i' \boldsymbol{\beta} \\ =& \varepsilon_i - \boldsymbol{x}_i' (\boldsymbol{b} - \boldsymbol{\beta}) \end{aligned}$$

Notes

Taking squares and doing some algebra

$$e_i^2 = \varepsilon_i^2 - 2(\boldsymbol{b}-\boldsymbol{\beta})'\boldsymbol{x}_i\varepsilon_i + (\boldsymbol{b}-\boldsymbol{\beta})'\boldsymbol{x}_i\boldsymbol{x}_i'(\boldsymbol{b}-\boldsymbol{\beta})$$

multiply both sides by n^{-1} and sum over n:

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^n e_i^2 =& \frac{1}{n}\sum_{i=1}^n \varepsilon_i^2 - 2(\boldsymbol{b}-\boldsymbol{\beta})'\frac{1}{n}\sum_{i=1}^n \boldsymbol{x}_i\varepsilon_i + (\boldsymbol{b}-\boldsymbol{\beta})'\left(\frac{1}{n}\sum_{i=1}^n \boldsymbol{x}_i\boldsymbol{x}_i'\right)(\boldsymbol{b}-\boldsymbol{\beta}) \\ =& \frac{1}{n}\sum_{i=1}^n \varepsilon_i^2 - 2(\boldsymbol{b}-\boldsymbol{\beta})'\overline{\boldsymbol{g}} + (\boldsymbol{b}-\boldsymbol{\beta})'\boldsymbol{S}_{\boldsymbol{xx}}(\boldsymbol{b}-\boldsymbol{\beta})\end{aligned}$$

Based on (B1)–(B4) we showed $\boldsymbol{b}-\boldsymbol{\beta} \xrightarrow{\mathsf{p}} \boldsymbol{0}$, hence

$$\frac{1}{n}\sum_{i=1}^n e_i^2 \xrightarrow{\mathsf{p}} \mathrm{E}[\varepsilon_i^2]$$

$$s^2 = \frac{n}{n-K}\left(\frac{1}{n}\sum_{i=1}^n e_i^2\right) \xrightarrow{\mathsf{p}} \mathrm{E}[\varepsilon_i^2]$$

■

Notes

Estimation of $\mathrm{AVar}(\boldsymbol{b})$

Asymptotic normality of $\sqrt{n}(\boldsymbol{b}-\boldsymbol{\beta})$ is central in order to construct inference. However, $\mathrm{AVar}(\boldsymbol{b})$ depends on $\boldsymbol{S}$ which is a population quantity.

Plugin method: replace $\boldsymbol{S}$ by some consistent estimator $\hat{\boldsymbol{S}}$:

$$\widehat{\mathrm{AVar}(\boldsymbol{b})} := \boldsymbol{S}_{\boldsymbol{xx}}^{-1}\hat{\boldsymbol{S}}\boldsymbol{S}_{\boldsymbol{xx}}^{-1}$$

where $\boldsymbol{S}_{\boldsymbol{xx}} := \frac{1}{n}\sum_{i=1}^n \boldsymbol{x}_i\boldsymbol{x}_i' = \frac{1}{n}\boldsymbol{X}'\boldsymbol{X}$

Proposition

Assume (B2) and assume there exists $\hat{\boldsymbol{S}} \xrightarrow{\mathsf{p}} \boldsymbol{S}$. Then as $n \to \infty$ $\widehat{\mathrm{AVar}(\boldsymbol{b})} \xrightarrow{\mathsf{p}} \mathrm{AVar}(\boldsymbol{b})$.

How to construct a reasonable $\hat{\boldsymbol{S}}$? When you don't know what to do, apply the analog principle (use empirical distribution) + plugin method (replace unknown by good estimates)

Notes

Use estimates and the empirical distribution to mimic $\boldsymbol{S} = \mathrm{E}[\varepsilon_i^2 \boldsymbol{x}_i \boldsymbol{x}_i']$:

$$\hat{\boldsymbol{S}} = \frac{1}{n}\sum_{i=1}^{n} e_i^2 \boldsymbol{x}_i \boldsymbol{x}_i' = \frac{1}{n}\boldsymbol{X}'\boldsymbol{B}\boldsymbol{X}$$

where

$$\boldsymbol{B} = \begin{pmatrix} e_1^2 & 0 & \cdots & 0 \\ 0 & e_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e_n^2 \end{pmatrix}$$

The asymptotic variance is easily calculated

$$\begin{aligned} \widehat{\mathrm{AVar}(\boldsymbol{b})} &= \boldsymbol{S}_{\boldsymbol{xx}}^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} e_i^2 \boldsymbol{x}_i \boldsymbol{x}_i'\right)\boldsymbol{S}_{\boldsymbol{xx}}^{-1} \\ &= n(\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{B}\boldsymbol{X})(\boldsymbol{X}'\boldsymbol{X})^{-1} \end{aligned}$$

Notes

Observe that

$$\mathrm{Var}\left[\sqrt{n}(\boldsymbol{b}-\boldsymbol{\beta})\right] = \mathrm{Var}[\sqrt{n}\boldsymbol{b}] = n\,\mathrm{Var}[\boldsymbol{b}]$$

$\sqrt{n}(\boldsymbol{b}-\boldsymbol{\beta}) \xrightarrow{\mathsf{d}} \mathcal{N}(\boldsymbol{0}, \mathrm{AVar}(\boldsymbol{b}))$, therefore as n grows sufficiently large the following asymptotic approximation holds

$$n\,\mathrm{Var}[\boldsymbol{b}] \overset{\mathsf{a}}{\approx} \mathrm{AVar}(\boldsymbol{b})$$

From which we compute an approximate standard error as follows

$$\begin{aligned} \mathrm{Se}(b_k) = \sqrt{\widehat{\mathrm{Var}}[b_k]} &= \sqrt{\frac{1}{n}\left(\widehat{\mathrm{AVar}(\boldsymbol{b})}\right)_{kk}} \\ &= \sqrt{\left((\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{B}\boldsymbol{X})(\boldsymbol{X}'\boldsymbol{X})^{-1}\right)_{kk}} \end{aligned}$$

Notes

Assumption (B6): existence of fourth moments

$\mathrm{E}[(x_{ik}x_{ij})^2]$ exists and it is finite for all $k, j = 1, 2, \ldots, K$.

(B6) implies that regressors are sampled from a population with *nice* kurtosis. It basically means *tails of the joint distribution of the regressors are not too fat*.

Proposition (consistency of $\hat{S}$)

Assume (B1), (B2), (B3), (B5), and (B6). Then $\hat{\boldsymbol{S}} \xrightarrow{\mathsf{p}} \boldsymbol{S}$ as $n \to \infty$.

Remark

asymptotic approximations often require high order moments (as for $\hat{\boldsymbol{S}}$), to rule out bad tail behaviors of the data generating process

Notes

Notes